Cambridge
O Level

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Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **October/November 2017**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)(i)	849 000;	**1**
1(a)(ii)	13.1(%);; *(if answer incorrect, allow one mark for 849 000 ÷ 6.5 [1])*;	**2**
1(a)(iii)	Knox, Sumner, Sullivan, Cumberland, Lauderdale; counties correctly paired with their population (441 000, 166 000, 157 000, 57 000, 28 000);	**2**
1(b)(i)	190; 118;	**2**
1(b)(ii)	*sulfate* 315 (ppm); *selenium* 60 (ppm);	**2**
1(b)(iii)	*stream* **B**;	**1**
1(b)(iv)	as pH increases pollutants decrease / eq;	**1**
1(b)(v)	(stream) **A**; *because:* lowest pH; highest value of, sulfate / aluminium / iron / manganese / selenium; AVP, e.g. calculated percentage compared to other streams or total ppm compared;	**3**
1(c)	*any three from:* selenium absorbed from water into plants / eq; then passed up the food chain; on to consumers; and top consumers; selenium cannot be excreted; so accumulates at each stage / bioaccumulates; so concentration, toxic / lethal to fish;	**3**

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Question	Answer	Marks
1(d)(i)	*any two from:* as a control experiment; so the data could be compared to polluted streams; so the degree / amount of pollution could be known;	**2**
1(d)(ii)	*allow answers in range of* 0.80–0.90 (m^2);; *(if answer incorrect, allow one mark for* (count up squares) x scale *[1])*;	**2**
1(d)(iii)	**B**; **A**;	**2**
1(d)(iv)	*any five from:* measure a, known / stated distance along stream; mark each end with a post; drop the float in upstream; start timer as it passes first marker; stop timer when it passes second marker; record time; repeat (at least twice more); distance ÷ time = speed / velocity;	**5**
1(d)(v)	*allow answers in range of* 0.144–0.162;; *(if answer incorrect, allow one mark for candidates answer to* ***(d)(ii)*** *× 0.18 [1])*;	**2**
1(e)	*any two advantages with one disadvantage OR any two disadvantages with one advantage:* *advantage:* limestone is cheap; easy / quick; prevents death of organisms; *disadvantage:* changes shape of stream / could cause flooding / eq; not a long term solution / needs to be repeated / requires labour / eq; alters, food chains / food webs / habitat; it does not remove pollutants / eq;	**3**

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Question	Answer	Marks
1(f)(i)	*any two advantages with one disadvantage OR any two disadvantages with one advantage:* *advantage:* a long term solution; does not add any chemicals; only uses natural processes / eq; no repeat treatments needed; few pollutants enter stream; *disadvantage:* high initial cost; damage to area in building / eq; only suitable for some streams / *ref to* topography; AVP, e.g. bacteria work best, at high temperatures / only during part of the year;	**3**
1(f)(ii)	*months* May, June, July, August, September; *reasons:* highest temperature means increased action of bacteria; (increased) enzyme activity; still plenty of water in wetland / eq so bacteria can live / eq;	**3**
1(f)(iii)	*any three from:* to check, that the system worked / pollution levels / eq; at highest rainfall (May) / beginning of the hot season; and lowest rainfall (Oct) / end of the hot season; to compare results for each year / between May and October; find out if water is safe to use;	**3**

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Question	Answer	Marks
2(a)(i)	coal output (steadily) decreased / use of figures to illustrate the fall in production, e.g. 2.3 to 1.1 / drop of 1.2;	**1**
2(a)(ii)	*any one from:* less coal available (to be mined) / eq; fall in demand / price; change to renewable sources; AVP;	**1**
2(b)(i)	orientation with linear scale; *axes labelled:* number of miners employed; year; plots correct;	**4**
2(b)(ii)	*allow answers in range of* 200–340;	**1**
2(b)(iii)	*any two from:* more machinery used; less coal to mine; less mines operating; miners move to other jobs, qualified;	**2**
2(c)	*any five from:* loss of natural landscape; deforestation; loss of habitats; loss of biodiversity; more erosion by wind; more erosion by water; increase risk of landslides; pollution / turbidity, of streams; pollution of groundwater; air pollution; dust reduces, plant growth / photosynthesis;	**5**

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Question	Answer	Marks
2(d)	*any four from:* *YES:* too much damage being done / high cost of restoration; people do not support it / no new licences; very few jobs; involves high cost of (long distance) transport; better things to spend state money on; most of the coal reserves have already been extracted; low quality coal not for burning; people want to use renewable energy sources; AVP; *NO:* valuable export; foreign exchange; earns money for the, state / country; provides jobs; subsidies keep mining going; fossil fuels still needed for, energy / industrial use; AVP;	**4**

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